

HONORS PHYSICS: KINEMATIC RENDEZVOUS CHALLENGE

HS-PS2-1 • SEP-5

Names: _____ Period: 0 Date: _____

UNIT 2: CONSTANT VELOCITY & GRAPHICAL INTERCEPT MODELING

MISSION SCENARIO: EMERGENCY HARBOR AIR-DROP

A marine research ship is traveling straight out to sea away from the harbor dock (dock = $x = 0\text{ m}$). At $t = 0\text{ s}$, the ship is already **300 meters** out from the dock, traveling at a steady speed of **6.0 m/s**.

An urgent medical distress call requires dispatching an autonomous courier drone from the dock ($x = 0\text{ m}$). The drone flies at a steady speed of **18.0 m/s**, but requires a **20-second preflight delay** (boot & rotor spin-up) before taking off at $t = 20\text{ s}$.

Research Vessel Telemetry

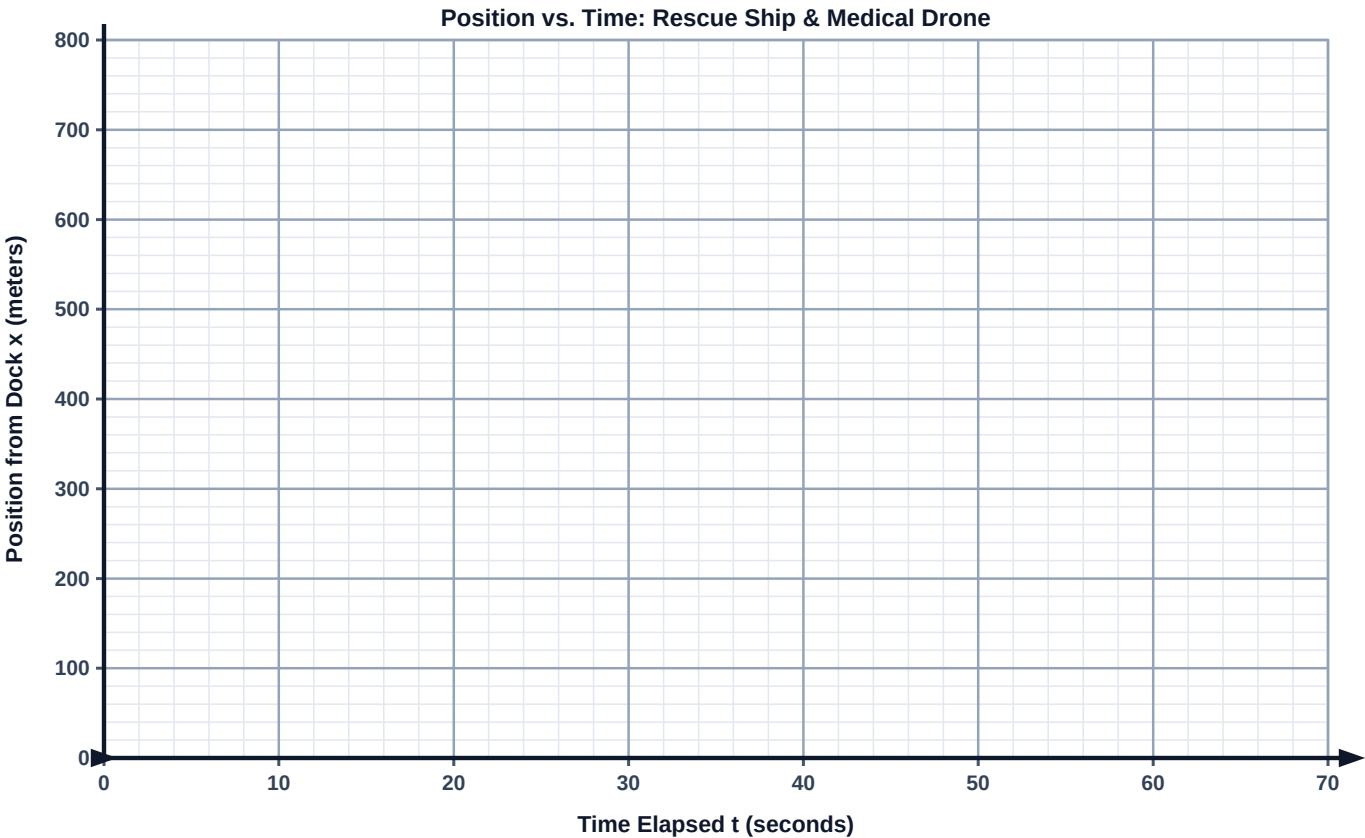
- Starts at:
 $x_0 = 300\text{ m}$
at $t = 0\text{ s}$
- Constant Speed:
 $v = 6.0\text{ m/s}$
(gains 60 m every 10 s)

Medical Drone Telemetry

- Starts at:
 $x_0 = 0\text{ m}$
(dock)
- Launch Delay: At rest until
 $t = 20\text{ s}$
, then cruises at
 $v = 18.0\text{ m/s}$
(gains 180 m every 10 s)

Graphing Task: Plot both motions on the grid. Label the Ship line and the Drone line. Circle the intersection point.

GRAPH-FIRST ANALYSIS



GRAPHICAL ANALYSIS & MISSION DEBRIEF

ANSWER USING YOUR GRAPH ABOVE

1. Rendezvous Coordinates (From Graph):

At what exact time and position do the two lines intersect? **Time (t):** _____ seconds **Position (x):** _____ meters from dock

2. Slope & Speed Verification:

Calculate the slope ($\Delta x / \Delta t$) of each graphed line to verify their speeds: Ship Slope: _____ m/s | Drone Slope (after $t = 20\text{ s}$): _____ m/s

3. Actual Drone Flight Duration:

The clock reads your rendezvous time when they meet, but the drone was delayed until $t = 20\text{ s}$. How many seconds was the drone actually flying in the air? _____ seconds

4. Battery Safety Audit (Honors Critical Question):

The drone has enough battery for a maximum round-trip flight distance of **1,200 meters** before complete depletion. Can the drone deliver the